

MATH 3210-004 Selected Solutions to PSet Problems

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Subject to Change; Last Updated: 2026-02-04 12:16:00-07:00

In this document you may find solutions (proofs or examples) to selected problems from problem sets. As a general rule of thumb, it is more likely that this document will be updated with the solutions to problems that were graded on accuracy.

Throughout this class your solutions need not match the level of polish, precision and detail of the below solutions. What's most important is grasping the underlying concepts and steadily improving your problem-solving skills as well as your proof-writing and presentation skills.

Most solutions will have follow-up exercises. Just as in the problem sets, these follow-up exercises are stated in a neutral language, and determining whether or not the statement is correct is part of the exercise. A follow-up exercise here may later show up in an exam.

1 PSet 1, Problem 3

Let X and Y be sets. Let R be a relation from X to Y . Then

- (i) The relation $\mathcal{R} = \{ (A, B) \mid B = R(A) \}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ is the graph of a function.
- (ii) For any $A, B \subseteq X$, $R(A \cup B) = R(A) \cup R(B)$.
- (iii) For any $A, B \subseteq X$, $R(A \cap B) = R(A) \cap R(B)$.
- (iv) For any $A \subseteq X$, $R(A^c) = R(A)^c$.
- (v) For any $A, B \subseteq X$, $R(A \setminus B) = R(A) \setminus R(B)$.

We'll study each statement separately. The first statement is true:

Claim: The relation $\mathcal{R} = \{ (A, B) \mid B = R(A) \}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ is the graph of a function. ┘

Proof: We need to verify that the relation $\mathcal{R}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ defined by

$$\mathcal{R} = \{(A, B) \mid B = R(A)\} \subseteq \mathcal{P}(X) \times \mathcal{P}(Y)$$

is the graph of a function with domain $\mathcal{P}(X)$. For $A \in \mathcal{P}(X)$, by definition $R(A) \in \mathcal{P}(Y)$, so that indeed $\text{dom}(\mathcal{R}) = \mathcal{P}(X)$. Next let $(A_1, B_1), (A_2, B_2) \in \mathcal{R}$. Suppose $A_1 = A_2$. We need to show that $B_1 = B_2$. As $(A_1, B_1), (A_2, B_2) \in \mathcal{R}$, by the definition of $\mathcal{R}$, we have that

$$B_1 = R(A_1), \quad B_2 = R(A_2).$$

To show the equality of two sets, it suffices to show that either one is a subset of the other one. So let $y \in B_1 = R(A_1)$. Then for some $x \in A_1$, $(x, y) \in R$. But as $A_1 = A_2$, $x \in A_2$ also, and since $(x, y) \in R$,

we have that $y \in R(A_2) = B_2$. Thus $B_1 \subseteq B_2$. Interchanging the roles of B_1 and B_2 in this argument, we immediately obtain also $B_1 \supseteq B_2$, whence $B_1 = B_2$. In other words, $\mathcal{R}$ satisfies the abstract vertical line test, and consequently is the graph of a function from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$.

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The second statement is also true:

Claim: For any $A, B \subseteq X$, $R(A \cup B) = R(A) \cup R(B)$.

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Proof: Let $A, B \subseteq X$. Then as $A \subseteq A \cup B$, $R(A) \subseteq R(A \cup B)$. Indeed, if $y \in R(A)$, there is an $x \in A$ such that $(x, y) \in R$. But $A \subseteq A \cup B$, so that $x \in A \cup B$, thus $y \in R(A \cup B)$. Similarly $R(B) \subseteq R(A \cup B)$. Then $R(A) \cup R(B) \subseteq R(A \cup B)$.

For the other direction, let $y \in R(A \cup B)$. Then there is an $x \in A \cup B$ such that $(x, y) \in R$. By definition $x \in A$ or $x \in B$, so that $y \in R(A)$ or $y \in R(B)$, that is, $R(A \cup B) \subseteq R(A) \cup R(B)$.

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The remaining three statements are false in general. To prove that a given statement is false, it suffices to provide one counterexample to the statement. For instance, the third statement is: "The image of the intersection of two subsets is equal to the intersection of the images". The **logical negation** of this is the statement "The image of the intersection of two subsets may be different than the intersection of the images". Using logical symbols, we may succinctly write:

$$\begin{aligned} & \neg[\forall A, B \in \mathcal{P}(X) : R(A \cap B) = R(A) \cap R(B)] \\ & \equiv \exists A, B \in \mathcal{P}(X) : R(A \cap B) \neq R(A) \cap R(B). \end{aligned}$$

We'll prove that the given statement is false, that is, the negation

of the given statement if true. As this latter is an existential statement, one example for it is sufficient; the example for the negation is a counterexample for the given statement.

Claim: The image of the intersection of two subsets under a relation is not necessarily equal to the intersection of the images.

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Proof: Consider $X = [0, 1] = Y$, $A = [0, 1/4]$, $B = [3/4, 1]$, $f : X \rightarrow Y$, $x \mapsto 1/2$, that is, f is the **constant function** that is constantly $1/2$. Then $A \cap B = \emptyset$, hence $f(A \cap B) = \emptyset$, while $f(A) = f(B) = \{1/2\}$, so that $f(A) \cap f(B) = \{1/2\}$.

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To obtain a correct version of the third statement, either we can weaken the conclusion, or strengthen the hypotheses¹.

Claim: The following version of the third statement with weakened conclusion is true:

$$\forall A, B \in \mathcal{P}(X) : R(A \cap B) \subseteq R(A) \cap R(B).$$

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Proof: Let $A, B \subseteq X$. As $A \cap B \subseteq A$, $R(A \cap B) \subseteq R(A)$. Similarly as $A \cap B \subseteq B$, $R(A \cap B) \subseteq R(B)$. Thus $R(A \cap B) \subseteq R(A) \cap R(B)$.

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Claim: The following version of the third statement with an additional hypothesis is true:

¹These are instances of logical **perturbations** of the original statement given; you do not need to submit such corrected versions, though trying to come up with such corrected versions will help you understand the concepts and prepare you for the exams better.

$$\begin{aligned} & [\forall (x_1, y_1), (x_2, y_2) \in R : y_1 = y_2 \implies x_1 = x_2] \\ \iff & [\forall A, B \in \mathcal{P}(X) : R(A \cap B) = R(A) \cap R(B)]. \end{aligned}$$

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Proof: ($\implies$) Suppose R satisfies

$$\forall (x_1, y_1), (x_2, y_2) \in R : y_1 = y_2 \implies x_1 = x_2;$$

one might call this the **injectivity condition** (aka **abstract horizontal line test**) for relations. Let $A, B \subseteq X$, and $y \in R(A) \cap R(B)$. Then there are $a \in A$ and $b \in B$ such that $(a, y), (b, y) \in R$. By the injectivity condition, this forces that $a = b$, and consequently that this element is in $A \cap B$. Thus $y \in R(A \cap B)$, so that $R(A) \cap R(B) \subseteq R(A \cap B)$. Since $R(A) \cap R(B) \supseteq R(A \cap B)$ for any relation R by the previous claim, we have $R(A) \cap R(B) = R(A \cap B)$.

($\impliedby$) Suppose R fails the injectivity condition, so that there are $x_1, x_2 \in X$ and $y \in Y$ such that $x_1 \neq x_2$ and $(x_1, y), (x_2, y) \in R$. Define $A = \{x_1\}$, $B = \{x_2\}$. Then $A \cap B = \emptyset$, so $R(A \cap B) = \emptyset$, but $y \in R(A) \cap R(B)$, that is, $R(A \cap B) \subsetneq R(A) \cap R(B)$.

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Claim: The image of the absolute complement of a subset under a relation need not be equal to the absolute complement of the image.

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Proof: We may repurpose our previous counterexample to establish this claim also. Put $X = Y = [0, 1]$, $A = [0, 1/4]$, $f : X \rightarrow Y$, $x \mapsto 1/2$. Then $A^c =]1/4, 1]$, $f(A^c) = \{1/2\}$, but $f(A)^c = \{1/2\}^c$, hence in fact we have that the two subsets are **disjoint**.

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Claim: The image of the relative complement of two subsets under a relation need not be equal to the relative complement of the images.

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Proof: Recalling that $A \setminus B = A \cap B^c$, the counterexample we gave earlier serves as a counterexample for this claim also. Indeed, put $X = Y = [0, 1]$, $A = [0, 1/4]$, $B = [3/4, 1]$, $f : X \rightarrow Y$, $x \mapsto 1/2$. Then $A \cap B^c = [0, 1/4]$, hence $f(A \setminus B) = \{1/2\}$; on the other hand $f(A) = f(B) = \{1/2\}$, so that $f(A) \setminus f(B) = \emptyset$.

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Follow-up Exercise 1 The second and the (correct versions) of the last third statements hold for arbitrary collections $\mathcal{A} \subseteq \mathcal{P}(X)$.

Follow-up Exercise 2 The fourth and fifth statement has correct versions (one involving a "subset of" relation, and one involving the injectivity condition; similar to the correct versions of the third statement).

Follow-up Exercise 3 If $f : X \rightarrow Y$ is a function, then the **preimage relation** f^{-1} satisfies the injectivity condition². Consequently, taking preimages **commutes** with unions, intersections and relative complements.

²Recall that f^{-1} really is an abbreviation for $\text{graph}(f)^{-1}$, and f^{-1} is not necessarily a function from Y to X .

2 PSet 1, Problem 6

Let X and Y be two sets. A function from X to Y is **surjunctive** if it is either surjective or non-injective. Then

- (i) X is finite iff any function from X to X is surjunctive.
- (ii) Y has exactly one element iff any function from X to Y is surjunctive.

The first claim is correct without any modifications, while the second statement is not quite correct as given. Still, a minor adjustment suffices to obtain a correct version. We start with the first claim.

For this claim, part of the challenge really is to make intuitive sense of what it means for a set to be finite³. From a set theoretical point of view, this calls for a **rigorous definition**, however we'll keep the notion of finiteness of sets at the naive level⁴. Thus for us, a set X is **finite** if it is either the emptyset or if it can be represented as

$$X = \{x_1, x_2, \dots, x_n\}$$

for some positive integer n . Similarly, a set X is **infinite** if for any positive integer n , there is an element $x_n \in X$ with the property that $x_i \neq x_j$ for $i \neq j$. With these definitions, for instance, it is true that saying that a set is not infinite is equivalent to saying that the set is finite, although proving this equivalence rigorously is surprisingly cumbersome! As we are really focused toward calculus, we'll not investigate this issue any further.

Claim: Let X be a set. Then X is finite iff for any $f \in F(X; X)$, f is

³In *this* sentence, we use the word "intuitive" intuitively and not in a **technical** sense. Throughout this course we'll almost never use "intuition" in a technical sense.

⁴Indeed, it is valid in the grand scheme of things to take the claim we set out to prove as the *definition* of what it means for a set to be finite.

surjunctive.

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Proof: Recall that "iff" stands for a necessary and sufficient condition. Thus we'll need to prove two implications.

($\Rightarrow$) First suppose $X = \{x_1, x_2, \dots, x_n\}$ is a finite set with exactly n elements and let $f : X \rightarrow X$ be an arbitrary function. We claim that f is surjunctive. That is, either f is surjective, or it is non-injective. To prove this, we may assume f fails to have one of these properties and show that it must have the other property. To that end, suppose f is injective (so that it fails to be non-injective). We need to show that f is surjective.

Suppose for a contradiction that there is a $j \in \{1, 2, \dots, n\}$ such that $x_j \notin \text{im}(f)$. But then $\text{im}(f)$ has at most $n - 1$ distinct elements. On the other hand, by the injectivity of f , for any $i, j \in \{1, 2, \dots, n\}$ with $i \neq j$, $f(x_i) \neq f(x_j)$, so that $\text{im}(f)$ has at least n distinct elements, a contradiction⁵.

($\Leftarrow$) For the **converse**, we'll argue contrapositively. Suppose X is infinite, so that we have for any positive integer n , an element $x_n \in X$ with the property that $x_i \neq x_j$ for $i \neq j$. We may then define a self-function of X as follows:

$$f : X \rightarrow X, x \mapsto \begin{cases} x_{n+1}, & \text{if } \exists n \in \mathbb{Z}_{\geq 1} : x = x_n \\ x, & \text{else} \end{cases}.$$

Thus f is the function that shifts the index of the specified indexed elements by one, and acts as identity on $X \setminus \{x_n | n \in \mathbb{Z}_{\geq 1}\}$. Then by construction, $x_1 \notin \text{im}(f)$, however f is injective, so that f fails to be surjunctive.

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⁵Alternatively, one can refer to the **pigeonhole principle** to derive a contradiction.

The second claim is not correct in general. Indeed, say $X = \{x_1, x_2, x_3\}$, $Y = \{y_1, y_2\}$. Then there are no injections from X to Y , thus any function $X \rightarrow Y$ is automatically surjunctive. The best we can do is to verify one side of the equivalence:

Claim: Let X and Y be two sets. If Y has exactly one element then any function $f : X \rightarrow Y$ is surjunctive.

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Proof: If $X = \emptyset$, then there are no functions $X \rightarrow Y$, so we may assume that $X \neq \emptyset$. Put $Y = \{y\}$ and let $f : X \rightarrow Y$ be function. Then f must be surjective, as there is at least one $x \in X$, and $f(x)$ is an element of Y , and consequently $f(x) = y$. Then by definition, f being surjective implies it being surjunctive.

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Follow-up Exercise Let Y be a set. Then Y has exactly one element iff for any set X , any function in $F(X; Y)$ is surjunctive.

3 PSet 2, Problem 1

$$\forall n \in \mathbb{Z}_{\geq 1} : \sum_{k=1}^n k^3 = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2.$$

The statement given is correct without any modifications. We'll give both an induction proof and a direct proof. For brevity let us first set up some notation. Put

$$S : \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}, (n, p) \mapsto \sum_{k=1}^n k^p.$$

Thus we have, for any n , $S(n, 0) = n$ and $S(n, 1) = \frac{n(n+1)}{2}$ (we proved this latter formula in class).

Claim:

$$\forall n \in \mathbb{Z}_{\geq 1} : S(n, 3) = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2.$$

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Proof (By Induction): For the base step, we need to verify that the statement holds for $n = 1$. Indeed; $S(1, 3) = 1$ and

$$\left[\frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2 \right] \Big|_{n=1} = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.$$

For the induction step, let $n_* \in \mathbb{Z}_{\geq 1}$ be arbitrary and suppose the statement holds for $n = n_*$, that is,

$$S(n_*, 3) = \frac{1}{4}n_*^4 + \frac{1}{2}n_*^3 + \frac{1}{4}n_*^2. \quad (*)$$

We need to show that the statement holds for $n = n_* + 1$. We'll start with the polynomial function on the RHS and through some algebraic

identities rewrite it in terms of $S(n_*, 3)$; specifically we'll make use of the **binomial formula**:

$$\begin{aligned}
 & \frac{1}{4}(n_* + 1)^4 + \frac{1}{2}(n_* + 1)^3 + \frac{1}{4}(n_* + 1)^2 \\
 &= \frac{1}{4}(n_*^4 + 4n_*^3 + 6n_*^2 + 4n_* + 1) \\
 & \quad + \frac{1}{2}(n_*^3 + 3n_*^2 + 3n_* + 1) \\
 & \quad + \frac{1}{4}(n_*^2 + 2n_* + 1) \\
 &= \left(\frac{1}{4}n_*^4 + \frac{1}{2}n_*^3 + \frac{1}{4}n_*^2 \right) \\
 & \quad + \left(\frac{4}{4} \right) n_*^3 \\
 & \quad + \left(\frac{6}{4} + \frac{3}{2} \right) n_*^2 \\
 & \quad + \left(\frac{4}{4} + \frac{3}{2} + \frac{2}{4} \right) n_* \\
 & \quad + \left(\frac{1}{4} + \frac{1}{2} + \frac{1}{4} \right) \\
 &\stackrel{*}{=} S(n_*, 3) + (n_*^3 + 3n_*^2 + 3n_* + 1) \\
 &= S(n_*, 3) + (n_* + 1)^3 \\
 &= S(n_* + 1, 3),
 \end{aligned}$$

where in the equality with $\star$ we used the induction hypothesis. Thus by the principle of mathematical induction, we are done.

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Proof (Direct Proof): For a direct proof, we'll relate the sum of p -th powers to the sums of lesser powers. Let $k, p \in \mathbb{Z}_{\geq 1}$ ⁶. Then

⁶When reading what follows, it might be helpful to take $p = 3$ to parse the manipulations.

$$(k+1)^{p+1} = \sum_{i=0}^{p+1} \binom{p+1}{i} k^i = k^{p+1} + (p+1)k^p + \sum_{i=0}^{p-1} \binom{p+1}{i} k^i$$

$$\Rightarrow k^p = \frac{(k+1)^{p+1} - k^{p+1}}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} k^i,$$

where for $\alpha, \beta \in \mathbb{Z}_{\geq 1}$ with $\alpha \geq \beta$,

$$\binom{\alpha}{\beta} = \frac{\alpha!}{\beta!(\alpha-\beta)!} = \frac{\alpha(\alpha-1)\cdots(\alpha-\beta+1)}{\beta(\beta-1)\cdots 2} \in \mathbb{Z}.$$

Hence for $n \in \mathbb{Z}_{\geq 1}$ we have

$$\begin{aligned} S(n, p) &= \sum_{k=1}^n k^p \\ &= \sum_{k=1}^n \left(\frac{(k+1)^{p+1} - k^{p+1}}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} k^i \right) \\ &= \frac{1}{p+1} \sum_{k=1}^n [(k+1)^{p+1} - k^{p+1}] - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} \left(\sum_{k=1}^n k^i \right) \\ &= \frac{1}{p+1} \sum_{k=1}^n [(k+1)^{p+1} - k^{p+1}] - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} S(n, i) \\ &= \frac{(n+1)^{p+1} - 1}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} S(n, i). \end{aligned}$$

We have that $S(n, 0) = n$ by definition. Thus plugging in $p = 1$ in this latter formula, we obtain

$$S(n, 1) = \frac{(n+1)^2 - 1}{2} - \frac{1}{2}n = \frac{n^2 + n}{2},$$

(thus giving another direct proof for the formula for the sum of first powers). Next we plug in $p = 2$ to obtain:

$$\begin{aligned}
 S(n, 2) &= \frac{(n+1)^3 - 1}{3} - \frac{1}{3} \sum_{i=0}^1 \binom{3}{i} S(n, i) \\
 &= \frac{(n+1)^3 - 1}{3} - \frac{1}{3} (S(n, 0) + 3S(n, 1)) \\
 &= \frac{(n+1)^3 - 1}{3} - \frac{1}{3} (S(n, 0) + 3S(n, 1)) \\
 &= \frac{(n+1)^3 - 1}{3} - \frac{1}{3} S(n, 0) - S(n, 1) \\
 &= \frac{(n+1)^3 - 1}{3} - \frac{n}{3} - \frac{n^2 + n}{2} \\
 &= \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.
 \end{aligned}$$

Finally taking $p = 3$ and substituting the formulas we obtained so far, we end up with the formula for the cubes of integers; we omit the routine algebraic manipulations.

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Note that while the direct proof seems longer and more complicated than the proof by induction, it does not require an explicit formula and hence proves more than what is claimed, and further it signifies a method for a direct proof for any p .

Follow-up Exercise 1 For any $p \in \mathbb{Z}_{\geq 1}$, $S(\bullet, p)$ is a **polynomial** of degree $p + 1$ with rational coefficients.

Follow-up Exercise 2 Let $\alpha(p), \mu(p)$ be the minimal numbers of rational number additions and rational number multiplications one would need to execute to compute all the coefficients of $S(\bullet, p)$, respectively, for p an arbitrary positive integer.

- (i) Compute $\alpha(p)$ and $\mu(p)$ for $p \in \{0, 1, 2\}$.
- (ii) Find two functions $\hat{\alpha}$ and $\hat{\mu}$ that can be written as the composition of finitely many **elementary functions** with the property that for some $C \in \mathbb{R}_{>0}$ and some $P \in \mathbb{Z}_{\geq 1}$:

$$\forall p \in \mathbb{Z}_{\geq P} : \quad \alpha(p) \leq C\hat{\alpha}(p), \quad \mu(p) \leq C\hat{\mu}(p).$$

- (iii) Find the **best** functions $\hat{\alpha}$ and $\hat{\mu}$ that have the properties in the previous part.

4 PSet 2, Problem 2

Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then

- (i) The set of all upper bounds of A is $[\sup(A), \infty[$.
- (ii) The set of all lower bounds of A is $] - \infty, \inf(A)]$.
- (iii) The smallest closed interval containing A is

$$[\inf(A), \sup(A)].$$

- (iv) The largest open interval contained in A is

$$] \inf(A), \sup(A)[.$$

Let $A \subseteq \mathbb{R}$. First let us set some notation for convenience. Define by $\mathcal{U}(A)$, $\mathcal{L}(A)$ the sets of all upper and lower bounds of A , respectively:

$$\mathcal{U}(A) = \{U \in \mathbb{R} \mid \forall a \in A : a \leq U\},$$

$$\mathcal{L}(A) = \{L \in \mathbb{R} \mid \forall a \in A : L \leq a\}.$$

We first establish that the first three statements are correct without modification:

Claim: Let $A \subseteq \mathbb{R}$ be nonempty and bounded, then

- (i) $\mathcal{U}(A) = [\sup(A), \infty[\subseteq \mathbb{R}$,
- (ii) $\mathcal{L}(A) =] - \infty, \inf(A)] \subseteq \mathbb{R}$,
- (iii) The smallest closed interval containing A is $[\inf(A), \sup(A)] \subseteq \mathbb{R}$.

- (iv) The largest open interval contained in A is $] \inf(A), \sup(A)[$ iff A is an **interval**.

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Proof: We are given that A is bounded and nonempty, hence we have that neither $\mathcal{U}(A)$ nor $\mathcal{L}(A)$ are empty. In particular, by the completeness axiom we have that the minimum of $\mathcal{U}(A)$ as well as the maximum of $\mathcal{L}(A)$ exist; $\sup(A)$ and $\inf(A)$ are *defined* to be these real numbers.

(i) and (ii) Note that if U is an upper bound, then any number larger than U too is an upper bound, and similarly one can replace a lower bound by an even lower number. Thus we have:

$$\mathcal{U}(A) = \bigcup_{U \in \mathcal{U}(A)} [U, \infty[, \quad \mathcal{L}(A) = \bigcup_{L \in \mathcal{L}(A)}] - \infty, L]. \quad (*)$$

In addition, if U is an upper bound of A , then $\sup(A) \leq U$, that is, $U \in [\sup(A), \infty[$, so that $\mathcal{U}(A) = [\sup(A), \infty[$. Similarly if L is a lower bound of A , L must be in $] - \infty, \inf(A)]$, so that $\mathcal{L}(A) \subseteq] - \infty, \inf(A)]$. This concludes the proof of the first two parts.

(iii) We may argue as follows for the third part: By definition $\inf(A)$ is a lower bound for any element in A and similarly $\sup(A)$ is an upper bound for any element in A , thus $A \subseteq [\inf(A), \sup(A)]$. If $[\lambda, \rho] \subseteq \mathbb{R}$ were a closed and bounded interval containing A , then λ would be a lower bound and ρ would be an upper bound for A ; hence by definition $\lambda \leq \inf(A)$ and $\sup(A) \leq \rho$. That is $[\inf(A), \sup(A)] \subseteq [\lambda, \rho]$. In other words, $[\inf(A), \sup(A)]$ indeed is the **smallest** closed interval containing A .

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Finally we study the last statement. In general the last statement

fails, indeed, for instance for $A = \{0, 1\}$, a set with exactly two elements, $1/2$ is in $]0, 1[=]\inf(A), \sup(A)[$ ⁷.

Still, if we assume that A is an interval to begin with, the fourth statement becomes true. Recall that the definition we are using for A to be an interval is that it satisfies the following property:

$$\forall x, y \in A, \forall t \in [0, 1] : (1 - t)x + ty \in A.$$

Claim: Let $A \subseteq \mathbb{R}$ be nonempty and bounded, then A is an interval iff the largest open interval contained in A is $] \inf(A), \sup(A)[$.

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Proof: Let A be nonempty and bounded. We have two cases: either $\inf(A) = \sup(A)$ xor $\inf(A) < \sup(A)$. In the first case, we have that A is a singleton, and consequently it does not contain any nonempty open interval, while it trivially satisfies the condition for being an interval, so that there is nothing to do. Suppose therefore that $\inf(A) < \sup(A)$.

($\Rightarrow$) Suppose A is a nonempty and bounded interval and let $\mu \in] \inf(A), \sup(A)[$. Then μ fails to be a lower bound as well as an upper bound of A , by the definitions of infimum and supremum, respectively. Thus there are $x, y \in A$ such that

$$\inf(A) \leq x < \mu < y \leq \sup(A).$$

Since A is an interval and $\mu = (1 - t_*)x + t_*y$ for $t_* = \frac{\mu - x}{y - x} \in]0, 1[$, $\mu \in A$. Therefore $] \inf(A), \sup(A)[\subseteq A$.

Let $] \lambda, \rho[$ be an arbitrary open interval contained in A . **WLOG** we may assume that this interval is nonempty and take $\mu \in] \lambda, \rho[$. Then $\mu \in A$ and hence $\mu \in [\inf(A), \sup(A)]$. It must be the case that $\mu > \inf(A)$, for otherwise, $\frac{\lambda + \inf(A)}{2}$, a number strictly less than $\inf(A)$, would

⁷For the purposes of problem set grading, concluding here would have been sufficient.

be in A , a contradiction. Similarly it must be the case that $\mu \leq \sup(A)$, whence $]\lambda, \rho[\subseteq]\inf(A), \sup(A)[$, so that indeed $] \inf(A), \sup(A)[$ is the largest open interval contained in A .

($\Leftarrow$) We'll argue by contradiction. Suppose A is bounded and nonempty, that $] \inf(A), \sup(A)[$, but that it fails to be an interval. Then there are $x, y \in A$ and $t_* \in]0, 1[$ such that $x \leq y$ and $(1 - t_*)x + t_*y \notin A$. But then

$$(1 - t_*)x + t_*y \in]x, y[\subseteq]\inf(A), \sup(A)[\subseteq A,$$

a contradiction. ┘

We make a final remark to contextualize this problem: We had discussed the **"clamp" heuristic for supremum and infimum in class**, and the problem really is a formalization of this heuristic: increasing L in $] - \infty, L]$ corresponds to tightening the lower clamp, decreasing U in $[U, \infty[$ corresponds to tightening the upper clamp, and the numbers L and U that correspond to the tightest possible configuration are the infimum and supremum, respectively. The heuristic works for clamping from without. While there are physical devices that **clamp from within**, they are different in principle than the abstract notions on infimum and supremum, whence the fourth part is incorrect unless we have additional information regarding the nature of the set we are trying to clamp down on.

5 PSet 2, Problem 3

Let $A, B \subseteq \mathbb{R}$. Then

$$(i) \sup(A \cup B) = \max\{\sup(A), \sup(B)\}$$

$$(ii) \inf(A \cup B) = \min\{\inf(A), \inf(B)\}.$$

We separate cases as usual: either at least one of A and B is empty, xor neither A nor B is empty.

The $A = \emptyset$ or $B = \emptyset$ Case WLOG, we may assume $A \neq \emptyset = B$. Then $A \cup B = A$ and

$$\sup(A \cup B) = \sup(A) > -\infty = \sup(\emptyset) = \sup(B),$$

$$\inf(A \cup B) = \inf(A) < \infty = \inf(\emptyset) = \inf(B).$$

Since the maximum of an extended real number R and $-\infty$ is R by definition, the first part follows. Similarly since the minimum of an extended real number R and ∞ is R by definition the second part is also true.

The $A \neq \emptyset \neq B$ Case For the first part, WLOG we may assume that $\sup(A) \geq \sup(B)$. Thus it suffices to show that $\sup(A) = \sup(A \cup B)$. First note that as $A \subseteq A \cup B$, by the **monotonicity** of supremum we have

$$-\infty \leq \sup(B) \leq \sup(A) \leq \sup(A \cup B) \leq \infty.$$

Either $A \cup B$ is unbounded from above, xor it is bounded from above. If $A \cup B$ is unbounded from above, then at least one of A or

B would have to be unbounded from above. If A is unbounded from above, then $\sup(A) = \infty = \sup(A \cup B)$ concludes the proof. If B is unbounded from above, then $\infty = \sup(B) \leq \sup(A) \leq \sup(A \cup B) \leq \infty$, hence again $\sup(A) = \sup(A \cup B)$ (and consequently A is unbounded from above also).

Now suppose $A \cup B$ is bounded from above; then the same is true for both A and B, and we have

$$-\infty \leq \sup(B) \leq \sup(A) \leq \sup(A \cup B) \leq \infty.$$

Say $\sup(A) < \sup(A \cup B)$. Then $\sup(A)$ is not an upper bound for $A \cup B$, so that there is an $x \in A \cup B$ such that

$$\sup(A) < x.$$

Since x exceeds the least upper bound for A, $x \notin A$, thus it must be the case that $x \in B$. But then

$$x \leq \sup(B) \leq \sup(A) < x;$$

a contradiction.

One can argue analogously for the statement involving infima. Alternatively one can argue as follows⁸:

$$\begin{aligned} \inf(A \cup B) &= -\sup(-(A \cup B)) \\ &= -\sup((-A) \cup (-B)) \\ &= -\max\{\sup(-A), \sup(-B)\} \\ &= -\max\{-\inf(A), -\inf(B)\} \\ &= \min\{\inf(A), \inf(B)\}, \end{aligned}$$

⁸You should check these steps if they are not clear!

where $-C = \{-c \in \mathbb{R} \mid c \in C\}$ for $C \subseteq \mathbb{R}$ a subset.

We have thus proved that indeed the statements in the problem are correct without any modifications:

Claim: $\forall A, B \subseteq \mathbb{R}$:

(i) $\sup(A \cup B) = \max\{\sup(A), \sup(B)\}$

(ii) $\inf(A \cup B) = \min\{\inf(A), \inf(B)\}.$

┘

Follow-up Exercise 1 $\forall \mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$:

(i) $\sup \left(\bigcup_{A \in \mathcal{A}} A \right) = \sup_{A \in \mathcal{A}} \sup(A),$

(ii) $\inf \left(\bigcup_{A \in \mathcal{A}} A \right) = \inf_{A \in \mathcal{A}} \inf(A).$

Follow-up Exercise 2 $\forall \mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$:

(i) $\sup \left(\bigcap_{A \in \mathcal{A}} A \right) = \inf_{A \in \mathcal{A}} \sup(A),$

(ii) $\inf \left(\bigcap_{A \in \mathcal{A}} A \right) = \sup_{A \in \mathcal{A}} \inf(A).$

6 PSet 3, Problem 2

Let $x_\bullet, y_\bullet$ be two convergent sequences of real numbers with limits $x_\infty, y_\infty \in \mathbb{R}$ respectively:

$$\lim_{n \rightarrow \infty} x_n = x_\infty, \quad \lim_{n \rightarrow \infty} y_n = y_\infty.$$

Then

- (i) $\lim_{n \rightarrow \infty} (x_n + y_n) = x_\infty + y_\infty$.
- (ii) For any real number α , $\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x_\infty$.
- (iii) $\lim_{n \rightarrow \infty} (x_n y_n) = x_\infty y_\infty$.
- (iv) $\lim_{n \rightarrow \infty} \max\{x_n, y_n\} = \max\{x_\infty, y_\infty\}$.

All these limit laws are indeed correct without any modifications. The last statement involving maxima is slightly trickier than the rest, so we'll study it separately.

Claim: Let $x_\bullet \rightarrow x_\infty, y_\bullet \rightarrow y_\infty$ be two convergent sequences of real numbers. Then

- (i) $\lim_{n \rightarrow \infty} (x_n + y_n) = x_\infty + y_\infty$.
- (ii) For any real number α , $\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x_\infty$.
- (iii) $\lim_{n \rightarrow \infty} (x_n y_n) = x_\infty y_\infty$.

┘

For statements like these, we'll need to take an arbitrary $\varepsilon \in \mathbb{R}_{>0}$ and find some finite time $N \in \mathbb{Z}_{\geq 0}$ such that after this time, the distance between the position at time n (n being a time later than N) and the claimed target at most ε . Perhaps surprisingly, instead of

starting with the ε , often it's more practical to start from the *end of the proof*. In other words, first with start with the distance we are trying to control, and then using some inequalities we establish that the distance we are trying to control can be controlled via the distances we are given control of. With all this said, a proof might go like this:

Proof: To be more concrete, let us start indexing the times of the sequences involved from 0⁹.

(i) For the first part, we have, for any n ,

$$|(x_n + y_n) - (x_\infty + y_\infty)| \leq |x_n - x_\infty| + |y_n - y_\infty| \quad (*)$$

by the triangular inequality. We know that each to the two summands on RHS can be made arbitrarily small, and so this will suffice.

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there are two times $N_1, N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N_1 \implies |x_n - x_\infty| < \varepsilon/2,$$

$$n \geq N_2 \implies |y_n - y_\infty| < \varepsilon/2.$$

But then, putting $N = \max\{N_1, N_2\}$, we have that for any $n \in \mathbb{Z}_{\geq 0}$, if $n \geq N$, then both of these bounds hold simultaneously. Thus for $n \geq N$, by (*), we have

$$|(x_n + y_n) - (x_\infty + y_\infty)| < \varepsilon.$$

(ii) We have for any $n \in \mathbb{Z}_{\geq 0}$,

⁹Ultimately at what time one starts indexing does not matter for the purposes of this proof, as the claim is all about the *future asymptotics*, insofar as the indices of different sequences are compatible.

$$|(\alpha x_n) - (\alpha x_\infty)| = |\alpha| |x_n - x_\infty| \leq (|\alpha| + 1) |x_n - x_\infty|. \quad (\dagger)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then $\varepsilon/(|\alpha| + 1) \in \mathbb{R}_{>0}$ also, and hence there is a time $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N \implies |x_n - x_\infty| < \varepsilon/(|\alpha| + 1).$$

Using this latter bound together with $(\dagger)$, we obtain that for any $n \geq N$,

$$|(\alpha x_n) - (\alpha x_\infty)| < \varepsilon.$$

(iii) This one is slightly more tricky. Whenever one tries to bound distances between "terms with factors", a good method is to "inject" some choice cross terms¹⁰. Thus we have, for any n ,

$$\begin{aligned} & |(x_n y_n) - (x_\infty y_\infty)| \\ &= |x_n y_n - x_n y_\infty + x_n y_\infty - x_\infty y_\infty| \\ &\leq |x_n y_n - x_n y_\infty| + |x_n y_\infty - x_\infty y_\infty| \\ &= |x_n| |y_n - y_\infty| + |x_n - x_\infty| |y_\infty| \\ &= |x_n - x_\infty + x_\infty| |y_n - y_\infty| + |x_n - x_\infty| |y_\infty| \\ &\leq (|x_n - x_\infty| + |x_\infty|) |y_n - y_\infty| + |x_n - x_\infty| |y_\infty|. \quad (\$) \end{aligned}$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then $\min\{1, \varepsilon/2(|y_\infty| + 1)\} \in \mathbb{R}_{>0}$ also¹¹, and since $x_\bullet \rightarrow x_\infty$, there is an $N_1 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

¹⁰Here by "inject" I mean adding and subtracting the same term and then using the triangular inequality.

¹¹As you are reading this proof, at this point you *should not* worry about why this number is relevant, and keep reading. For now, this is merely a possible error margin that depends on ε already chosen. See also the next footnote at the end of the proof.

$$n \geq N_1 \implies |x_n - x_\infty| < \min\{1, \varepsilon/2(|y_\infty| + 1)\}.$$

Further, $\varepsilon/2(|x_\infty| + 1) \in \mathbb{R}_{>0}$ too, and since $y_\bullet \rightarrow y_\infty$, there is an $N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N_2 \implies |y_n - y_\infty| < \varepsilon/2(|x_\infty| + 1).$$

Combining these two inequalities with (\$), we have that, putting $N = \max\{N_1, N_2\}$, for any $n \in \mathbb{Z}_{\geq 0}$,

$$n \geq N \implies |(x_n y_n) - (x_\infty y_\infty)| < \varepsilon.^{12}$$

┘

Claim ((iv)): Let $x_\bullet \rightarrow x_\infty, y_\bullet \rightarrow y_\infty$ be two convergent sequences of real numbers. Then

$$\lim_{n \rightarrow \infty} \max\{x_n, y_n\} = \max\{x_\infty, y_\infty\}.$$

┘

We'll provide two separate proofs of this claim; one is more straight-

¹²How did I come up with the error margins for $x_\bullet$ and $y_\bullet$, as functions of ε , so that when the dust settled I obtained the nice ε bound for what I ultimately wanted to control? You'll see such bizarre expressions in the middle of proofs in analysis quite often; and the answer almost always is this: the author of the proof again worked *backwards*, and did some calculations **not worthy of explanation** (supposedly). In such cases it's best to not dwell too much on the specific choices of numbers and to keep reading the proof (as mentioned in the previous footnote). For instance in this case I was really looking at the RHS in (\$). I want this term to be less than ε . As this is the sum of two terms, I decided to simply bound both summands by $\varepsilon/2$ each. The second summand has a factor that may or may not be 0, so I scaled $\varepsilon/2$ by dividing it by $|y_\infty| + 1 \in \mathbb{R}_{>0}$. But then I realized I'm going to need to bound the first term by $\varepsilon/2$ also, and this scaled version of $\varepsilon/2$ is too cumbersome to manipulate algebraically, hence I simply took the minimum of it with 1, in case the first supposed bound is relatively large etc.. There are many other bounds that would have worked, and ultimately none of this is really necessary. Instead you can simply bound the RHS by some function of ε that decays to zero as ε does. For instance in this case it's much more straightforward to bound the RHS by a quadratic polynomial in ε that has a zero at $\varepsilon = 0$ (such a polynomial decays like ε^2 , which for ε small enough, is even smaller than ε). There are **contexts** where one *needs* to be careful about sharp bounds (for instance if the problem was asking statements regarding the **rates of convergences**), but often there is a fair amount of flexibility with epsilon management.

forward but requires that one first establishes an inequality regarding maxima, while the other one is less straightforward but does not use any technology. For the first proof we first establish a lemma:

Lemma: For any $\alpha, \bar{\alpha}, \beta, \bar{\beta} \in \mathbb{R}$,

$$|\max\{\alpha, \bar{\alpha}\} - \max\{\beta, \bar{\beta}\}| \leq \max\{|\alpha - \beta|, |\bar{\alpha} - \bar{\beta}|\}.$$

┘

Proof: Wlog, we may assume $\alpha \leq \bar{\alpha}$ and $\beta \leq \bar{\beta}$. Then

$$|\max\{\alpha, \bar{\alpha}\} - \max\{\beta, \bar{\beta}\}| = |\bar{\alpha} - \bar{\beta}| \leq \max\{|\alpha - \beta|, |\bar{\alpha} - \bar{\beta}|\}.$$

┘

Proof (of Claim (iv), using Lemma): We have that, by the lemma, for any $n \in \mathbb{Z}_{\geq 0}$:

$$|\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| \leq \max\{|x_n - x_\infty|, |y_n - y_\infty|\}. \quad (\spadesuit)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there are $N_1, N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N_1 \implies |x_n - x_\infty| < \varepsilon,$$

$$n \geq N_2 \implies |y_n - y_\infty| < \varepsilon.$$

Consequently, putting $N = \max\{N_1, N_2\}$, by $(\spadesuit)$, we have that for any $n \in \mathbb{Z}_{\geq 0}$,

$$n \geq N \implies |\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| < \varepsilon.$$

┘

Proof (of Claim (iv), without using Lemma): Wlog, we may assume that $x_\infty \leq y_\infty$. Then either $x_\infty \leq y_\infty$ xor $x_\infty = y_\infty$.

The $x_\infty \leq y_\infty$ Case In this case we have that $(y_\infty - x_\infty)/2 \in \mathbb{R}_{>0}$. Thus, using this positive number as the error margin, we have that there is a time $N \in \mathbb{Z}_{\geq 0}$ such that¹³ for any $n \in \mathbb{Z}_{\geq 0}$,

$$\begin{aligned} n \geq N \\ \implies |x_n - x_\infty| &< (y_\infty - x_\infty)/2 \\ \text{and } |y_n - y_\infty| &< (y_\infty - x_\infty)/2. \end{aligned}$$

This implies¹⁴ that

$$n \geq N \implies x_n \leq y_n.$$

Consequently, if $n \geq N$, we have

$$|\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| = |y_n - y_\infty|. \quad (\clubsuit)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there is an $N' \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$,

$$n \geq N' \implies |y_n - y_\infty| < \varepsilon.$$

Putting $N'' = \max\{N, N'\}$, using this last inequality together with $(\clubsuit)$, we have that for any $n \in \mathbb{Z}_{\geq 0}$,

¹³Here we are fast-forwarding the procedure of first finding two separate times for the two sequences to be sufficiently near their respective limits and then taking the maximum of these two times so as to guarantee simultaneity.

¹⁴Draw a picture, or rearrange the above inequalities!

$$n \geq N'' \implies |\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| < \varepsilon.$$

The $x_\infty = y_\infty$ Case For clarity denote by L the common limit. Then both $x_\bullet$ and $y_\bullet$ converges to L . Let $\varepsilon \in \mathbb{R}_{>0}$. Then there is an $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq N}$,

$$|x_n - L| < \varepsilon,$$

$$|y_n - L| < \varepsilon.$$

Rearranging these inequalities, we obtain that for any $n \in \mathbb{Z}_{\geq N}$,

$$L - \varepsilon < x_n < L + \varepsilon,$$

$$L - \varepsilon < y_n < L + \varepsilon.$$

Note that the upperbound the maximum of two numbers, one needs to upperbound *both numbers involved*. Hence the RHS's of the inequalities above give us

$$\forall n \in \mathbb{Z}_{\geq N} : \max\{x_n, y_n\} < L + \varepsilon.$$

To lowerbound the maximum of two numbers, one needs to lowerbound *at least one of the numbers involved*. The LHS's of the inequalities above in fact are *more than enough* to conclude

$$\forall n \in \mathbb{Z}_{\geq N} : L - \varepsilon < \max\{x_n, y_n\}.$$

Combining these two last inequalities, we have:

$$\forall n \in \mathbb{Z}_{\geq N} : |\max\{x_n, y_n\} - L| < \varepsilon.$$

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Follow-up Exercise 1 Here is one way to get more quantitative versions of the above results: Let $x_\bullet : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ be a convergent sequence. The limit of $x_\bullet$ is a uniquely determined real number, so we don't need to explicitly specify it. Then for any $\varepsilon \in \mathbb{R}_{>0}$, the following set of non-negative integers is nonempty:

$$Z_\varepsilon = \left\{ N \in \mathbb{Z}_{\geq 0} \mid \forall n \in \mathbb{Z}_{\geq N} : \left| x_n - \lim_{n \rightarrow \infty} x_n \right| < \varepsilon \right\}.$$

Then by the Well-Ordering Principle, this set has a unique minimum; let's denote this minimum by $\mathcal{N}_\varepsilon(x_\bullet)$. Thus $\mathcal{N}_\varepsilon(x_\bullet)$ is the *exactly* how long one would need to wait for $x_\bullet$ to be within an ε of its limit. We can extend this definition to arbitrary sequences by setting $\mathcal{N}_\varepsilon(x_\bullet) = \infty$ if $Z_\varepsilon = \emptyset$, so that ultimately we have a function

$$\mathcal{N} : \mathbb{R}_{>0} \times F(\mathbb{Z}_{\geq 0}; \mathbb{R}) \rightarrow \mathbb{Z}_{\geq 0} \cup \{\infty\}, (\varepsilon, x_\bullet) \mapsto \mathcal{N}_\varepsilon(x_\bullet).$$

Then the above statements can be rephrased in terms of $\mathcal{N}$. For instance, the proof of the statement regarding the limit of the sums of two sequences can be rephrased as:

$$\forall \varepsilon \in \mathbb{R}_{>0} : \mathcal{N}_\varepsilon(x_\bullet + y_\bullet) \leq \max\{\mathcal{N}_{\varepsilon/2}(x_\bullet), \mathcal{N}_{\varepsilon/2}(y_\bullet)\}.$$

How about an *exact* formula for the time on LHS, for this part or the other parts?

Follow-up Exercise 2 One can **generalize** the lemma above as follows: Let X be a set, $\alpha, \beta \in F(X; \mathbb{R})$. Then

$$|\sup \alpha - \sup \beta| \leq \sup |\alpha - \beta|.$$

Follow-up Exercise 3 State and prove versions of the last statement with minima.

7 PSet 3, Problem 3

Let $x_\bullet$ be a sequence of real numbers that converges to zero.
Then for any bounded sequence $b_\bullet$ of real numbers,

$$\lim_{n \rightarrow \infty} x_n b_n = 0.$$

This claim is indeed true without any modifications:

Claim: $\forall x_\bullet, b_\bullet : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R} : \text{if } x_\bullet \rightarrow 0 \text{ and } \sup_m b_m < \infty, \text{ then } x_\bullet b_\bullet \rightarrow 0.$ ┘

Proof: Instead of starting with an $\epsilon > 0$, it's often useful to start from the end, in other words with the expression we are trying to bound. For any $n \in \mathbb{Z}_{\geq 0}$, one has

$$|x_n b_n| = |x_n| |b_n| \leq |x_n| \underbrace{\sup_{m \in \mathbb{Z}_{\geq 0}} |b_m|}_{< \infty} \quad (*)$$

Put $M = \sup_{m \in \mathbb{Z}_{\geq 0}} |b_m| + 1 \in \mathbb{R}_{\geq 0}$. We know that this is a finite nonnegative number, as the sequence $b_\bullet$ (and consequently the sequence $|b_\bullet|$) is bounded. We added 1 to the supremum of $|b_\bullet|$ to ensure that M is a positive number, so that dividing by M is well-defined¹⁵. Thus to bound $|x_n b_n|$ from above by ϵ , we can bound $|x_n|$ by ϵ/M . Indeed, let $\epsilon \in \mathbb{R}_{>0}$. Then as $x_\bullet \rightarrow 0$, there is an $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq N}$, we have

$$|x_n| = |x_n - 0| < \frac{\epsilon}{M}.$$

¹⁵Alternatively, we could have done a case analysis: either $b_\bullet$ is the sequence that is constantly zero, in which case there is nothing to do, or else $b_\bullet$ is not constantly zero, so that the supremum of the absolute value sequence $|b_\bullet|$ is strictly positive.

Here we used the number $\frac{\varepsilon}{M}$ as the "epsilon" in the definition of x_n converging to 0. For any $n \in \mathbb{Z}_{\geq N}$, $(*)$ also holds; hence combining the two we have that for any $n \in \mathbb{Z}_{\geq N}$:

$$|x_n b_n - 0| = |x_n b_n| < \frac{\varepsilon}{M}(M - 1) < \varepsilon.$$

┘

Follow-up Exercise Let x_n be a sequence of real numbers. Then the following are equivalent:

- (i) x_n converges to 0.
- (ii) For any bounded sequence b_n , $x_n b_n$ converges to 0.
- (iii) For some bounded sequence b_n , $x_n b_n$ converges to 0.